# SPRINGBOARD VIRTUAL INTERNSHIP 6.0

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**College:** Sona College of Technology

**Branch:** Computer Science and Engineering

**Domain:** Data Visualization

**Project:** U.S. Natural Disaster Declarations Analysis

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## Milestone-02: Temporal Trend Analysis

### 1. Objective

The objective of this milestone is to analyze how disaster declarations change over time using the FEMA dataset. The analysis focuses on identifying yearly trends, comparing disaster types, and observing seasonal patterns through visualizations.

### 2. Time Data Validation

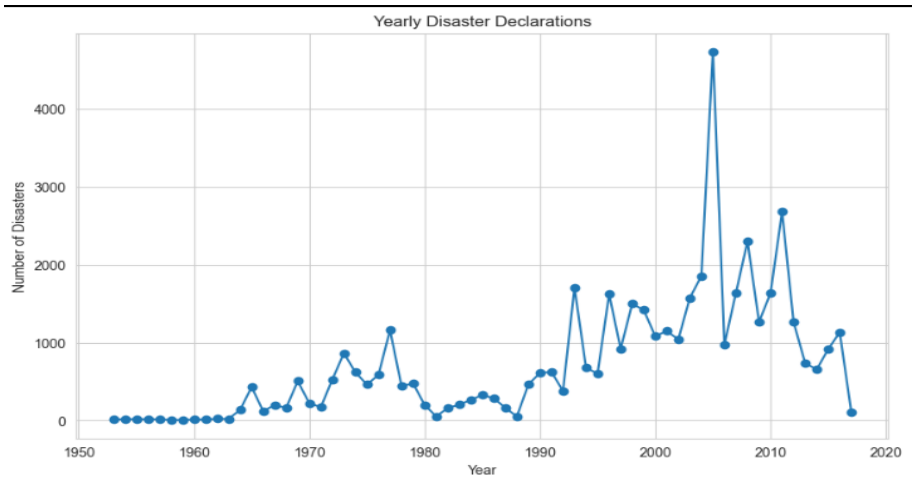
The declaration date column was converted to datetime format. From this column, **year** and **month** were extracted for time-based analysis. The dataset was also checked for missing or invalid date values.

### 3. Yearly Trend Analysis

The data was grouped by **year** to calculate the number of disaster declarations per year. A line chart was used to visualize the yearly trend.

**Insight**

- The graph shows the yearly distribution of disasters.
- A gradual increase in disaster declarations can be observed.
- Some years show sudden spikes indicating higher disaster activity.

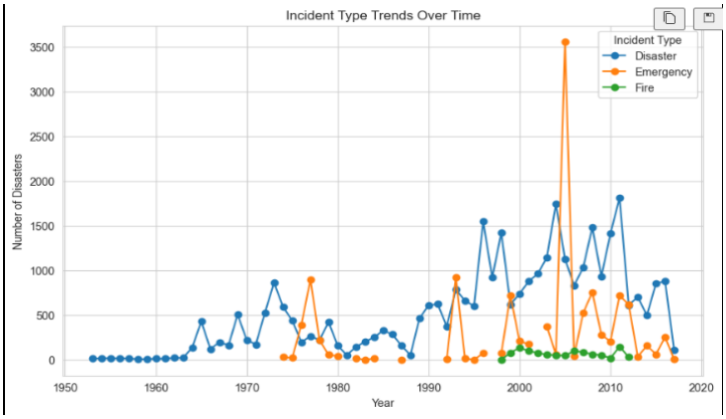


## 4. Incident Type Trends

The dataset was grouped by **year and incident type** to compare disaster categories over time. A multi-line chart was used for visualization.

**Insight**

- The chart compares multiple disaster types across years.
- Certain disaster types occur more frequently than others.
- Some categories show increasing trends.

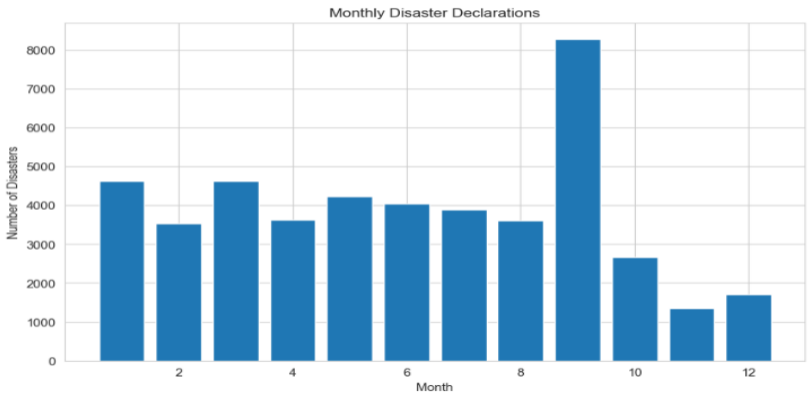



## 5. Seasonality Analysis

Disaster declarations were grouped by **month** to study seasonal patterns. A bar chart was used to visualize the monthly distribution.

**Insight**

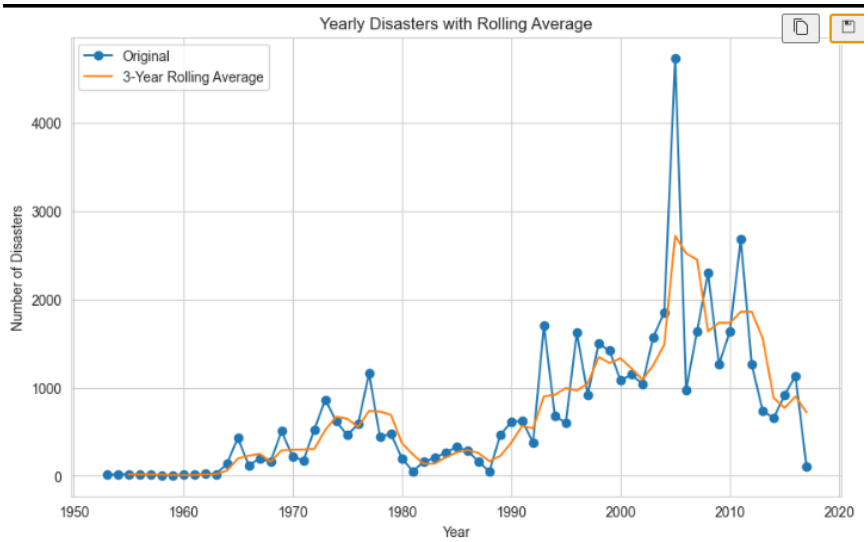
- Some months show higher disaster activity.
- This indicates seasonal influence on disaster occurrences.



## 6. Advanced Insights

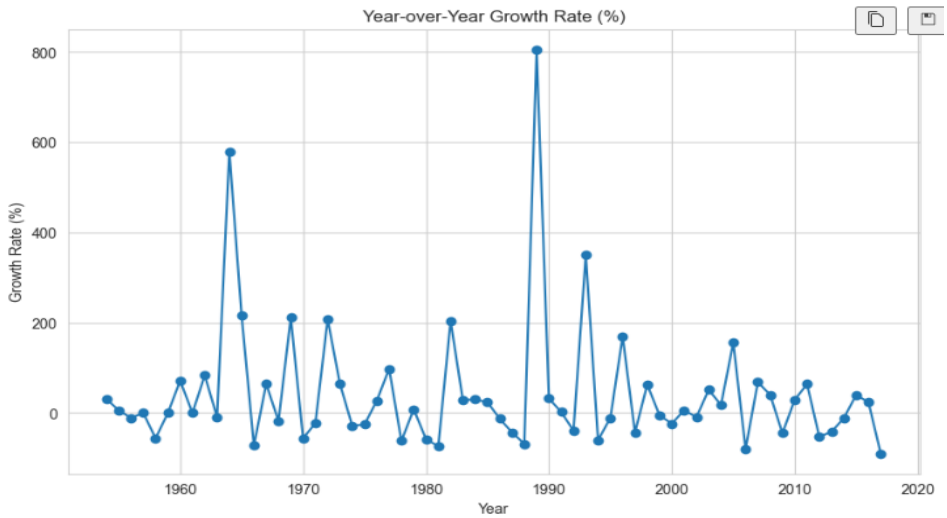
### Rolling Average

A rolling average was applied to smooth yearly variations and highlight long-term trends.




### Growth Rate

Year-over-year growth rate was calculated to observe changes in disaster declaration increase or decrease over time.



### Peak Year

The year with the highest number of disasters was identified.

## 7. Conclusion

The temporal analysis shows variations in disaster declarations across years and months. The results also highlight differences among disaster types and seasonal patterns in disaster occurrences